

Prior-Guided Gaussian Splatting for High-Fidelity 4D Expression, Head Pose, and Gaze Capture

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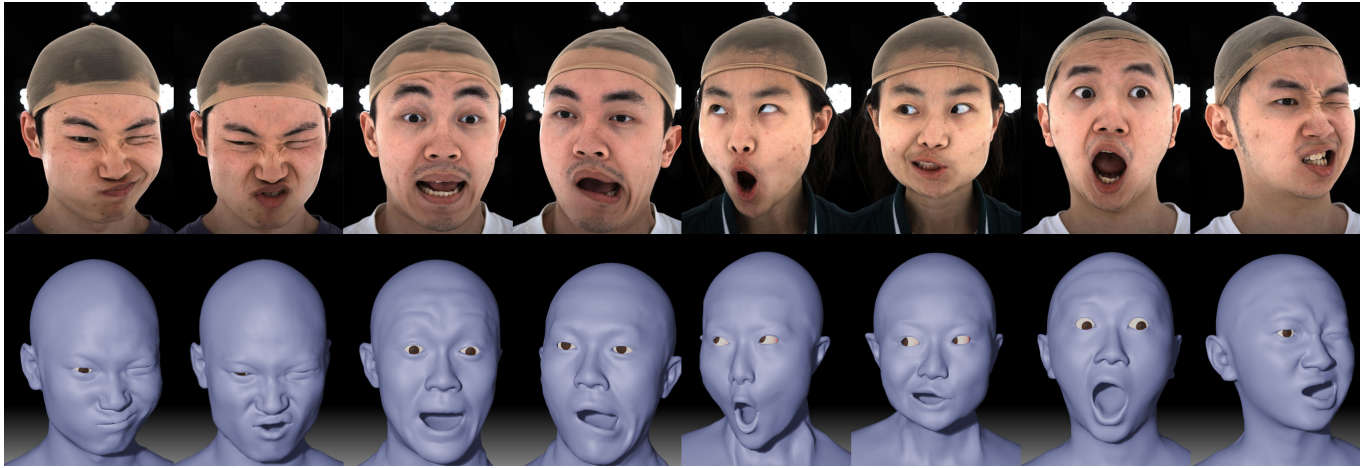


Fig. 1. Given calibrated multi-view videos, our scan-free 4D reconstruction method can faithfully capture the performer's head pose, expression and gaze.

Production-level digital humans require not only emotional facial expression dynamics but also realistic head pose and gaze to achieve lifelike presence. However, traditional capture pipelines are costly to reconstruct facial details and struggle to acquire pose and gaze simultaneously. These pipelines rely

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on complex workflows involving specialized operation, high computational overhead, and labor-intensive post-processing, making them unsuitable for large-scale production. To address these limitations, we introduce a scan-free 4D reconstruction framework that simultaneously captures topology-consistent expression dynamics, pose, and gaze from calibrated multi-view videos. Our approach decomposes the task into a prior-guided multi-stage tracking process. By integrating multi-source priors at varying granularities, including coarse model-based priors for initialization and fine-grained Gaussian-derived priors for refinement, we effectively reduce cumulative errors and mutual interference among pose, expression, and gaze. Furthermore, to capture the geometry details and address the over-smoothing issue, we design a collaborative strategy between a topology-preserving Gaussian set and a detailed non-topological Gaussian set, where the former provides reliable structural and motion priors while the latter captures details. Based on this hybrid representation, we propose a normal-guided detail transfer mechanism that effectively enhances the mesh details. Comprehensive experiments demonstrate that our method outperforms existing approaches in reconstruction quality and robustness.

CCS Concepts: • **Computing methodologies** → **Reconstruction; Motion capture.**

Additional Key Words and Phrases: 4D Head Capture, Gaze Tracking, 4D Gaussian Splatting

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1 INTRODUCTION

Faithfully capturing 4D topology-consistent head assets is a fundamental enabler for creating vivid human avatars widely used in entertainment media, such as movies and games. A production-level digital human head with immersive quality goes beyond covering realistic and detailed facial expression dynamics, but also expresses believable head features, including arbitrary head poses and eye gazes. However, the complexity of facial muscle movements, limited observation of eyeballs, cumulative errors during prolonged capture sessions, and mutual errors among expression, pose, and gaze collectively make the 4D full-head capture task exceptionally challenging.

In current industrial practice, the capture of detailed facial expression dynamics, head pose, and eye gaze is typically handled through separate pipelines to reduce system complexity. To produce 4D facial expressions, high-precision multi-view videos are captured by professional equipment, e.g., Light Stage [Cao et al. 2018; Debevec 2012; Debevec et al. 2000; Vlasic et al. 2009]. Then, the reconstruction process typically involves two time-consuming and labour-intensive steps. First, the multi-view stereo (MVS) method is employed to reconstruct high-precision but noisy scans from multi-view captures. Subsequently, a non-rigid tracking and registration method aligns the face mesh template with fixed topology to the scans [Beeler et al. 2011; Besl and McKay 1992; Rusinkiewicz and Levoy 2001; Wu et al. 2018]. Despite some methods [Fyffe et al. 2017] can directly wrap a template mesh to multi-view images without relying on scan input, they require time-consuming mesh surface/volume resolving processes. To ensure semantic and temporal coherence of the reconstructed mesh under extreme expressions or in the presence of capture noise, considerable manual intervention is required, including parameter tuning and artist-led post-processing. Additionally, subjects are typically instructed to minimize head movements to prevent degradation in expression reconstruction. For high-precision head pose capture, motion capture remains the industry standard. This is commonly implemented with systems that utilize either an array of reflective markers tracked by cameras or inertial measurement units to derive the head pose [Lee and Joo 2024]. Meanwhile, accurate eye motion capture [Bérard et al. 2019, 2014; Garbin et al. 2020; Schwartz et al. 2020] often relies on head-mounted eye trackers, where active IR illumination-based methods are used to intrusively determine the pose of the eyes from spectral features of corneal reflections. The aforementioned traditional pipelines are commonly confronted with the complexities of device calibration and operation, further complicated by the cumbersome integration of separately tracked pose, eye, and face. Therefore, there is an urgent demand to develop an automatic system to simultaneously reconstruct detailed facial expression dynamics, head pose, and 3D eye gaze from captured RGB images.

In recent years, numerous methods have been proposed to automate the capture of face [Biwen et al. 2023; Chai et al. 2023; Yoon et al. 2019; Zhang et al. 2022; Zoss et al. 2019, 2020], head pose [Li et al. 2020; Liu et al. 2020; Yu et al. 2018; Zhang et al. 2023], and eye motion [Bérard et al. 2016; Krafka et al. 2016; Smith et al. 2013;

Zhang et al. 2020] using deep learning, optimization, or hybrid strategies. To reduce task complexity, many of these approaches incorporate facial priors to enhance robustness and fidelity. However, due to the limitations of oversimplified facial representations and insufficient facial priors, previous methods struggle to achieve high-precision joint capture, as illustrated in Fig.2 (a). A prominent line of research [Feng et al. 2021; Taubner et al. 2024; Wood et al. 2022; Zielonka et al. 2022] builds on parametric models [Blaž and Vetter 1999; Li et al. 2017; Paysan et al. 2009], which leverage learned priors to compactly represent facial attributes. Despite their practicality, these models often rely on low-dimensional, entangled parameters to encode identity and expression, limiting their ability to recover high-frequency facial details. As a result, they are primarily suitable for low-precision scenarios, typically with monocular input [Bai et al. 2023; Thies et al. 2016; Zollhöfer et al. 2018]. Another class of methods [Bolkart et al. 2023; Li et al. 2024b, 2021; Liu et al. 2022] directly regresses topological meshes from multi-view images. However, these approaches generally demand large amounts of expensive annotated data for training. They consequently encode the training set-specific priors and are significantly affected by the data bias, making them susceptible to overfitting and hindering generalization to unseen expressions or diverse capture conditions. To achieve high-fidelity facial reconstruction, optimization-based [Beeler et al. 2011; Bradley et al. 2010; Fyffe et al. 2017; Smith et al. 2020; Yang et al. 2020] methods are commonly employed to register a template mesh to scan or multi-view images. Most of these methods use facial landmarks, optical flow, or other facial prior constraints to guide the registration process. However, such priors offer only coarse and ambiguous cues about facial structure and motion. Moreover, the mesh deformation process tends to be computationally expensive, slow, and highly sensitive to hyperparameter settings. Some recent scan-free methods bypass the scan reconstruction step by deforming a topological mesh directly to match image observations, which inevitably require more accurate and denser priors as guidance. The parametric methods [Qian 2024; Zielonka et al. 2022] leverage robust priors from parametric models for monocular or multi-view head capture. Although these methods typically incorporate further vertex-level optimization, they remain constrained by the inherent limitations of the underlying parametric models. In contrast, some methods achieve vertex-level dense reconstruction without dependence on parametric priors. Topo4D [Cheng et al. 2025; Li et al. 2024a] introduces an innovative topology-aware 3D Gaussian representation, termed Gaussian Mesh, which leverages topological and physical priors to track facial pixels. Despite its flexibility, the method is prone to geometric inaccuracies, particularly in texture-less regions where RGB images offer insufficient constraints. The absence of scan-based geometric supervision further forces a trade-off between mesh over-smoothness and topological integrity, resulting in loss of fine details. Additionally, as the capture session lengthens, error accumulation can lead to tracking instability or failure.

When capturing head pose, eye gaze, and facial expression jointly, a central challenge arises from their highly interdependent nature, which leads to mutual interference and the propagation of errors. Some parametric model-based approaches [Qian 2024; Zhang et al. 2023] support joint capture through costly pre-defined mesh joints

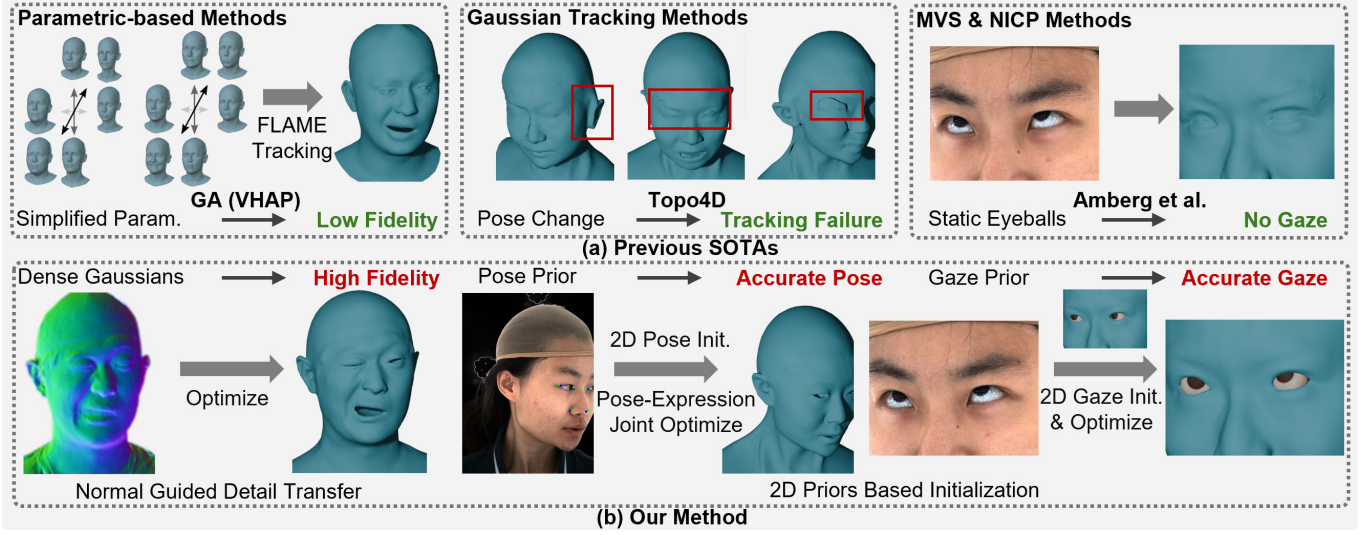


Fig. 2. Comparison between existing state-of-the-art face reconstruction methods and our method. (a) Previous SOTAs struggle to simultaneously capture high-precision expression, head pose and gaze due to the limitation of rough priors and simplified representations. (b) Our method effectively achieves faithful joint capture through diverse prior guidance and normal-guided detail transfer mechanisms.

or blendshapes, yet their limited expressiveness often leads to noticeable inaccuracies in each output dimension, despite the overall plausibility afforded by strong parametric facial priors. Although Topo4D [Cheng et al. 2025; Li et al. 2024a] models a more accurate facial expression dynamics via vertex-level dense tracking, directly extending it to capture pose and gaze remains non-trivial. This is due to a circular dependency: detailed expression reconstruction depends on precise head pose, while reliable pose and gaze tracking, in turn, requires well-recovered facial expressions. When performing vertex-level dense optimization, mutual errors among the three components become more pronounced and can even cause complete capture failure. One potential solution is to initialize pose and gaze using 2D estimators [Albiero et al. 2021; Bao and Lu 2024; Cheng et al. 2022; Hempel et al. 2022, 2024; Ververas et al. 2024]. However, existing generalizable models are highly sensitive to variations in capture environment and lighting, limiting their generalization across capture setups. Moreover, propagating such noisy estimations into subsequent dense optimization stages still fails to ensure robustness. Therefore, a more integrated and stable optimization strategy is needed to resolve this interdependency.

As illustrated in Fig. 2 (b), to overcome the aforementioned challenges, we present a prior-guided 3D Gaussian tracking framework that benefits from varying facial priors and hybrid Gaussian head representations, enabling the joint accurate capture of extreme facial expression dynamics, head pose, and 3D gaze from multi-view RGB videos without requiring scan input. Specifically, 1) we design a multi-stage optimization framework that explicitly accounts for the dependencies and disparities among topological face modeling, geometric detail recovery, head pose, and gaze tracking. Each stage leverages intermediate results obtained from preceding steps to guide subsequent optimization. For instance, recovering the facial geometric details requires a semantically aligned coarse mesh, while

gaze capture benefits from accurate expression and pose. By staging the optimization processes appropriately, our framework facilitates mutual reinforcement across different tracking tasks and supports high-quality joint tracking. 2) To tackle the over-smoothing issue common in existing scan-free methods, we introduce a normal-guided mesh refinement strategy. Our observation is that surface normal maps provide rich geometric cues and can be efficiently optimized by frame-by-frame incremental optimization, unlike costly scan reconstruction. By maintaining a detailed non-topological Gaussian set alongside a coarse topological Gaussian mesh, we transfer high-frequency details between the hybrid head representations via normal consistency without breaking mesh topology. 3) In addition, to address the optimization instability arising from the ambiguity of photometric objectives in head pose and gaze optimization, we propose a novel strategy of rough prior-guided initialization followed by dense prior-guided refinement to fully leverage the complementary strengths of multi-dimensional priors. Specifically, we first initialize these parameters using coarse but generalizable priors derived from off-the-shelf 2D facial landmark detectors [Lugaresi et al. 2019] and gaze estimators [Ververas et al. 2024]. These initial estimations are then jointly refined together with facial expressions using fine-grained Gaussian-derived geometry and appearance priors, thereby effectively alleviating the optimization instability caused by the error propagation and cyclic dependencies of joint capture. Extensive experiments show that our method achieves superior geometric accuracy than existing state-of-the-art scan-free face reconstruction methods, and is robust to extreme expressions, arbitrary head pose and gaze, as shown in Fig. 1. In summary, our key contributions are as follows:

- We present a multi-stage optimization framework that simultaneously tracks detailed facial expression dynamics, head

pose, and eye gaze by tightly integrating topological and non-topological 4D face reconstruction. This integration enables mutual enhancement between the joint capture processes, effectively mitigating cumulative and mutual errors.

- We introduce a normal-guided detail transfer method that enriches the topological Gaussian face with high-frequency geometric details, thereby effectively alleviating the over-smoothing problem commonly found in scan-free methods.
- We propose a prior-guided head and gaze tracking method to effectively enhance the robustness of joint tracking, which leverages multi-source facial priors at varying granularities, including 2D model-based priors, mesh topology priors, as well as Gaussian-derived geometry and appearance priors.

2 RELATED WORK

2.1 4D Head Capture

Efficiently reconstructing topologically consistent and temporally stable dynamic head assets has long been a challenging problem. The current industrial pipeline typically first reconstructs the sequence of scans captured by a professional device through the multi-view stereo (MVS). Then, a template face mesh will be registered to the scans by non-rigid ICP [Deng et al. 2022; Martinez et al. 2024; Wu et al. 2023], with manual modifications from artists. To simplify and automate such an extremely costly process, early works use keypoints or optical flow to guide mesh deformation, but these methods are sensitive to hyperparameters and still cannot get rid of the troublesome scan reconstruction. Many later works attempt to directly regress topological facial geometry from monocular or multi-view images [Bolkart et al. 2023; Cao et al. 2015; Guo et al. 2020; Riviere et al. 2020; Wang et al. 2024c; Zhu et al. 2017]. Parametric models [Bianz and Vetter 1999; Chandran et al. 2020; Egger et al. 2020; Li et al. 2017] have become the core of most of these methods due to their ability to efficiently represent faces using low-dimensional parameters. However, parametric faces fail to reconstruct high-precision facial details, especially extreme facial expressions and ID-related facial features. Many multi-view works strive to move beyond the limitations of parametric models by directly optimizing vertex-level deformations of the facial mesh. The first category [Bolkart et al. 2023; Li et al. 2021; Liu et al. 2022] encompasses learning-based approaches that directly regress vertex positions from multi-view images. However, these methods require a large amount of expensive registered training data, and struggle to generalize to unseen extreme expressions and other capture systems. The second category involves optimization-based methods that eschew scans as geometric supervision, relying primarily on photometric constraints [Fyffe et al. 2017]. Among these, GaussianAvatars (VHAP) [Qian 2024; Qian et al. 2024] starts from a parametric model but employs a perturbation mechanism to optimize static and dynamic vertex deformations to enhance fidelity. The method most related to ours is Topo4D [Cheng et al. 2025; Li et al. 2024a]. Topo4D utilizes a topology-aware Gaussian mesh, binding 3D Gaussians to mesh vertices and achieving reconstruction through Gaussian motion tracking [Luiten et al. 2024]. However, the concurrent demands of motion tracking and topology preserving result in mesh

over-smoothing and loss of detail. In contrast, our method simultaneously reconstructs a dynamic, high-fidelity non-topological head alongside tracking the topological face. The facial details can be transferred under normal guidance without breaking the topology, effectively resolving the issue of mesh over-smoothing.

2.2 Head Pose and Gaze Tracking

Beyond high-fidelity facial expressions, accurate head pose and eye gaze are critical facial features for enhancing the realism of the head. Directly optimizing for precise pose, expression, and gaze simultaneously is challenging due to their inherent interdependency, often leading to significant cumulative and mutual errors. Consequently, most existing approaches decouple the tracking of pose, expression, and gaze. Accurate pose-based facial alignment is fundamental for subsequent reconstruction. To obtain precise head poses, one category of methods directly estimates head rotation angles from images. Early techniques [DeMenthon and Davis 1995] employed a two-step process involving facial landmark detection followed by PnP algorithms [Lepetit et al. 2009]. However, these landmark-based approaches exhibit substantial errors under partial facial visibility or extreme expressions. Recent methods eliminate the need for landmark detection or face recognition, yielding 3DoF [Ruiz et al. 2018; Yang et al. 2019] or 6DoF [Albiero et al. 2021; Deng et al. 2020; Hempel et al. 2024] poses from images. Nevertheless, the inherent error in these image-based estimations remains insufficient for high-precision facial reconstruction. An alternative category [Li et al. 2020, 2017; Zhang et al. 2023] parametrically models head rotation jointly with expression, optimizing pose parameters or blending weights. However, these methods often suffer from limited expressiveness due to a simple face model or require pre-defined skeletons or blendshapes.

For gaze tracking, traditional methods employ intrusive head-mounted devices [Mackworth and Thomas 1962] relying on infrared illumination to capture gaze [Yu et al. 2015]. This process involves cumbersome equipment setup and calibration, hindering widespread adoption. Subsequent learning-based methods [Kellnhofer et al. 2019; Zhang et al. 2017] regress gaze direction directly from images. Yet, due to the high correlation of gaze with eyelid shape and illumination, these methods typically exhibit poor generalization. While recent approaches aim for generalizable in-the-wild estimation, they still incur significant errors and struggle with alignment to facial meshes. Some 3D methods [Liang et al. 2022; Wang et al. 2016; Wen et al. 2020, 2017, 2016; Wood et al. 2018] jointly solve gaze tracking and face/eyelid reconstruction, often requiring depth input or relying on separately trained models to estimate facial/ocular landmarks or eyelid contours. The facial features are then combined with 3D model coordinates to infer gaze direction or used to match parametric eye models. These techniques remain susceptible to errors from 2D detectors and are often limited by simplified expression models, resulting in low fidelity. Distinct from these, our method avoids simplistic parametric representations and leverages model-based priors to reduce the optimization ambiguity of joint pose, vertex-level facial deformation, and gaze tracking.

2.3 Gaussian-Based Geometry Reconstruction

Following the success of 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] in novel view synthesis, numerous works leverage its capability for efficient and photorealistic rendering to reconstruct scenes and extract detailed geometry. However, the inherent random and disordered discrete representation of 3DGS makes direct surface extraction challenging. Early attempts [Guédon and Lepetit 2024; Huang et al. 2024; Yu et al. 2024] design surface regularization terms or replace 3D ellipsoids with 2D planes to address multi-view inconsistency, aiming to better align Gaussians with surfaces. Yet, these methods remain reliant on biased depth, hindering high-fidelity reconstruction. Subsequent works [Chen et al. 2024; Zhang et al. 2024] improve Gaussian rasterizers by rendering unbiased depth and normal maps, enabling detailed scene reconstruction under multi-view geometric consistency constraints. A lot of research extends the 3DGS to reconstruct dynamic scenes. The global optimization [Wang et al. 2024b; Wu et al. 2024; Yang et al. 2024] methods typically learn deformation fields for Gaussian surfaces across all frames, jointly optimizing dynamic Gaussian distributions. Conversely, incremental optimization [Chen et al. 2025; Zheng et al. 2024] approaches guide the deformation of Gaussian surfaces using priors like optical flow. Unlike these methods, our approach utilises a coarse topological Gaussian head to guide the reconstruction of a detailed non-topological Gaussian head. Furthermore, we leverage unbiased normal maps rendered from the non-topological detailed surface to enhance detail on the topological Gaussian surface. This coarse-to-fine process between the hybrid head representations enables the extraction of a topological high-fidelity mesh from the Gaussian set.

3 FRAMEWORK OVERVIEW

Our method aims to track high-fidelity facial expressions, head poses, and 3D eyeball gazes from calibrated multi-view videos. Specifically, given a set of temporally synchronized videos $\{\mathbf{I}_t^c \in \mathbb{R}^{h \times w \times 3} | 0 \leq t < F\}_{c=0}^{C-1}$ captured by a Light Stage from C views, and each video encompasses F RGB frames in the resolution of $h \times w$ with known camera intrinsic matrix $\mathbf{K}_c \in \mathbb{R}^{3 \times 3}$ and extrinsic matrix $\mathbf{W}_c \in \mathbb{R}^{4 \times 4}$. Our method can track the motion of the initial head mesh $\mathbf{M}_0$, and generate the sequence of posed head meshes $\{\mathbf{M}_t := (\mathbf{V}_t^M, \mathbf{E}_M) | \mathbf{V}_t^M \in \mathbb{R}^{n_v^M \times 3}, \mathbf{E}_M \in \mathbb{R}^{n_e^M \times 2}\}_{t=0}^{F-1}$ and head-aligned left and right eyeballs $\{\mathbf{B}_t^{\{L,R\}} := (\mathbf{V}_t^{\{L,R\}}, \mathbf{E}_B) | \mathbf{V}_t^{\{L,R\}} \in \mathbb{R}^{n_v^B \times 3}, \mathbf{E}_B \in \mathbb{R}^{n_e^B \times 2}\}_{t=0}^{F-1}$, where the topology of the head and eyeball are fixed, with $n_v^{\{M,B\}}$ vertices and $n_e^{\{M,B\}}$ edges respectively.

Our method operates in a frame-by-frame manner, and the reconstruction of each frame coordinates four optimization stages to resolve the interdependence between detailed facial geometry, head pose, and gaze. Notably, to avoid the computational expense of the traditional scan reconstruction process while achieving high precision reconstruction, our method works without requiring scan inputs or reliance on any parametric face models. Due to the lack of scan as explicit geometric supervision, existing scan-free methods commonly confront the problem of over-smoothing, resulting in the loss of geometric details. The main reasons leading to this issue

are (1) the geometric information in RGB images is ambiguous, especially in texture-less facial areas, and (2) to maintain topological stability during reconstruction, it is inevitable to apply smoothing constraints to balance mesh quality and geometric accuracy. To solve the above problems, our idea is to use two representations with different granularities to represent dynamic heads. We simultaneously optimize a coarse but topology-preserving head and a high-precision but non-topological head, with the former focusing on maintaining topological and semantical stability and the latter preserving high-precision geometry and appearance details. By utilizing the normal of the high-precision head, we can transfer details to the topological head without breaking the topology, and the following gaze tracking can also benefit from the accurate facial geometry. The overall framework of our method is demonstrated by Fig. 3. To begin with, we briefly summarize the key components involved in our tracking framework as follows.

Head and Eyeball Representation. Following Topo4D [Li et al. 2024a], we represent the topological coarse head and eyeball with a *Gaussian Mesh*. Taking head as an example, the coarse head at frame t is represented by a set of 3D Gaussian points $G_t^M = \{G_{t,i}^M\}_{i=0}^{n_v^M-1}$, where each Gaussian point is represented by some learnable parameters, i.e., $\{\mu_{t,i}^M \in \mathbb{R}^3, \mathbf{q}_{t,i}^M \in \mathbb{R}^4, \mathbf{s}_{t,i}^M \in \mathbb{R}^3, \mathbf{c}_{t,i}^M \in \mathbb{R}^3, \sigma_{t,i}^M \in \mathbb{R}\}$ for the mean position, rotation represented by quaternion, scaling, RGB color, and opacity of i -th 3D Gaussian point. These parameters describe a Gaussian distribution:

$$G(\mathbf{x}) = e^{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)}, \quad (1)$$

where the covariance matrix $\Sigma = \mathbf{R} \mathbf{S} \mathbf{S}^T \mathbf{R}^T$ is further decomposed into rotation matrix $\mathbf{R}$, parameterized with $\mathbf{q}$, and scaling matrix $\mathbf{S}$. In a Gaussian mesh, the position of each 3D Gaussian point stands for a mesh vertex of the topological head, and the Gaussian points are related by the topological structure $\mathbf{E}_M$. The Gaussian mesh can be rendered like a vanilla Gaussian set:

$$C = \sum_{i=1} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (2)$$

where the color C of a pixel is acquired by blending all Gaussians that splats to the pixel in depth order, and the blending weight α is calculated by evaluating the splatted 2D Gaussian with its covariance multiplied by its opacity. The Gaussian mesh $G_t^{\{L,R\}}$ of the eyeballs can be formulated similarly.

We use a separate Gaussian set $\hat{G}_t^M$ to represent a higher precision head that focuses on preserving details. Unlike the topological head G_t^M , $\hat{G}_t^M$ does not maintain a fixed topological relationship, and behaves like a vanilla Gaussian set. The initialization and optimization of the two Gaussian heads are described in Sec. 4.1 and Sec. 4.2.

Topology-Preserving Coarse Mesh Tracking. Our method tracks the head frame by frame. The first step of each frame is to track the topological Gaussian Mesh G_t^M to obtain a coarse but semantically aligned head mesh. Following Topo4D [Li et al. 2024a], by introducing topology-aware physical and topological prior-based constraints, the mesh can roughly align with multi-view images and maintain the semantic consistency of the facial region. However, when the head pose is not fixed, the Gaussian points tend to lose

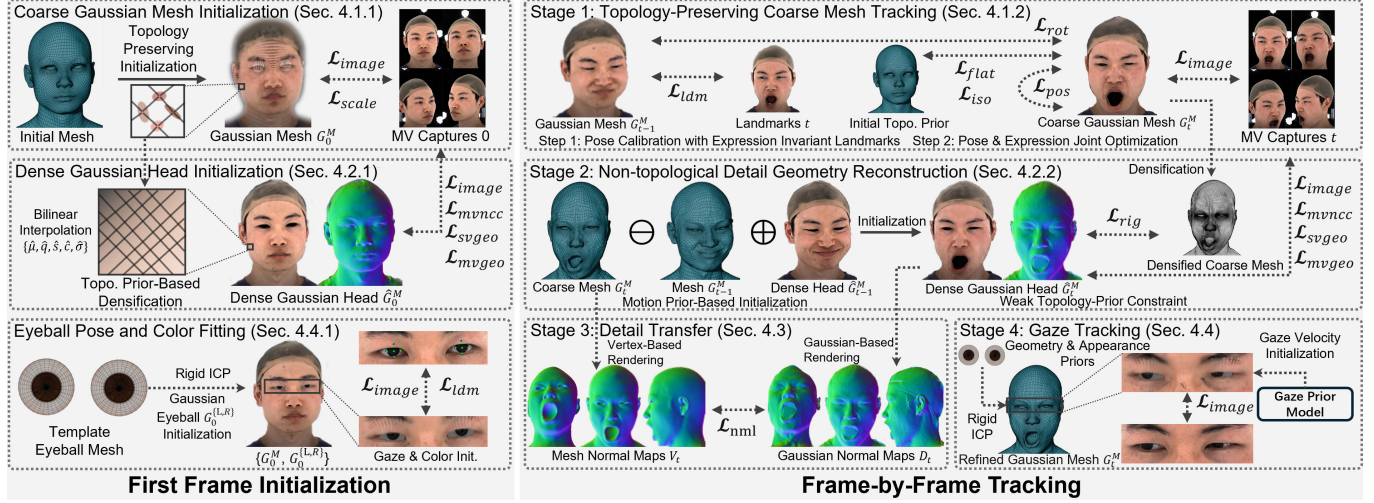


Fig. 3. Overview of our joint pose-expression-gaze tracking method. Starting from an initial mesh, we leverage its topological and geometric priors to construct a Gaussian mesh (Sec. 4.1.1) and a dense Gaussian head (Sec. 4.2.1), optimizing the initial gaze and eyeball appearance (Sec. 4.4.1). The tracking contains 4 stages. (a) In stage 1 (Sec. 4.1.2), we perform pose alignment using expression-invariant landmarks and then jointly refine the pose and expression dynamics while preserving the topology. (b) In stage 2 (Sec. 4.2.2), we utilize the motion and topology priors from the coarse head to reconstruct the detailed non-topological head. (c) In stage 3 (Sec. 4.3), geometric details are transferred to the coarse mesh via a normal consistency constraint, preserving the underlying topology. (d) Finally in stage 4 (Sec. 4.4), we initialize the gaze from an off-the-shelf prior model and then refine it via the facial geometry and appearance priors.

tracking due to large global motion. We solve this problem by introducing a facial landmark-guided pose calibration process and then jointly optimizing the head pose and expression dynamics.

Non-topological Detail Geometry Reconstruction. Based on the coarse head mesh of the current frame, we then optimize the non-topological head $\hat{G}_t^M$. Considering that learning detailed geometry needs denser Gaussian points with finer granularity than G_t^M , we hierarchically initialize $\hat{G}_t^M$ by densifying G_t^M based on its topology structure. To reduce the difficulty of tracking the dense $\hat{G}_t^M$, we utilize the motion prior from G_t^M to initialize the motion of $\hat{G}_t^M$ and further apply the geometric consistency losses together with a weak topology-prior constraint term to incrementally optimize the detailed Gaussian head.

Geometric Detail Transfer. We propose a normal-guided detail transfer method to refine the coarse head G_t^M without breaking its topology by deforming the vertices supervised by the rendered multi-view normals from $\hat{G}_t^M$ (Sec. 4.3). We also utilize the unbiased normal maps to refine the head pose, bridging the gap between the Gaussian surface and the real surface, further improving the geometric accuracy.

Gaze Tracking. After acquiring accurate facial geometry, we conduct gaze tracking. However, directly optimizing gaze from images remains challenging due to the eyeball's limited observable area and consequent appearance ambiguity. Considering that facial expression and gaze are closely associated, we integrate the Gaussian head into the gaze tracking process, thereby fully utilizing the reliable geometric and appearance information derived from the accurately reconstructed expression of the current frame. Furthermore, to ensure robustness, we employ a generalizable gaze estimation model

that provides a high-quality initial state by estimating the gaze velocity.

4 METHODOLOGY

In this section, we dive into the details of the key components involved in our framework, especially the initialization and optimization strategies for tracking a high-fidelity head with gaze, as summarized in Algorithm 1.

4.1 Tracking Posed Coarse Gaussian Head

4.1.1 Coarse Gaussian Mesh Initialization. Our method begins with a textured topological neutral head mesh of the current performer from the first frame. This can be readily acquired through a landmark-based non-rigid ICP followed by a multi-view texture mapping process. We first initialize the Gaussian mean positions μ_0^M of the Gaussian head G_0^M by rigging the Gaussian points to the mesh vertex, i.e., $\mu_0^M := V_0^M$. Here, we ignore the vertex ID subscript and only write the timestep index for simplicity. The quaternions q_0^M are initialized by the mesh vertex normal direction, and the Gaussian colors c_0^M are set to the vertex color sampling from the texture. To achieve a close correspondence between Gaussian points and facial regions, we set the opacity σ_0^M to 1, the initial scales s_{init}^M as half of the minimum distance between each Gaussian point and its one-ring neighbours defined by the topology E_M .

Based on the accurate color and geometry initialization, we only optimize s_0^M and q_0^M by the image loss:

$$\mathcal{L}_{image} = (1 - \lambda_{image}) \mathcal{L}_1(I_t^c, \tilde{I}_t^c) + \lambda_{image} \mathcal{L}_{D-SSIM}(I_t^c, \tilde{I}_t^c), \quad (3)$$

where $\tilde{I}_t^c$ is the image rendered by the Gaussian rasterizer, and $t = 0$ here. To align Gaussian points more closely with the face surface

Algorithm 1: Facial Dynamics, Head Pose and Gaze Capture

Input: Multi-view captures $\{I_{0:F-1}^c\}_{c=0}^{C-1}$ from C views, camera intrinsics K_c and extrinsics W_c , head mesh of the first frame $M_0 = (V_0^M, E_M)$, template eyeball $B_0^{\{L,R\}} = (V_0^{\{L,R\}}, E_B)$.

Output: 4D head meshes $M_{0:F-1}$ and eyeball meshes $B_{0:F-1}^{\{L,R\}}$.

// First frame initialization.

```

1  $G_0^M \leftarrow \text{Bind\_Gaussian}(M_0);$  // Build Gaussian Mesh
2  $G_0^M \leftarrow \arg \min_{G^M} \mathcal{L}_{init};$  // Init. coarse Gaussian
3  $\hat{G}_0^M \leftarrow \text{Densify}(G_0^M);$ 
4  $\hat{G}_0^M \leftarrow \arg \min_{\hat{G}^M} \mathcal{L}_{detail};$  // Init. dense Gaussian
5  $G_0^{\{L,R\}} \leftarrow \text{Bind\_Gaussian}(B_0^{\{L,R\}});$ 
6  $G_0^{\{L,R\}} \leftarrow \text{ICP}(G_0^M, G_0^{\{L,R\}});$  // Bind eyeball
7  $(G_0^{\{L,R\}}, r_0^{\{L,R\}}, v_0^{\{L,R\}}) \leftarrow \text{Fit\_Pos\_Color}(G_0^{\{L,R\}}, I_0^{0:C-1});$ 
8 for  $t = 1$  to  $F - 1$  do // Frame by frame tracking
    // Stage 1: Topology-Preserving Coarse Mesh Tracking
    9  $(r_t^M, v_t^M) \leftarrow \arg \min_{r^M, v^M} \mathcal{L}_{ldm};$  // Init. pose
    10  $(G_t^M, r_t^M, v_t^M) \leftarrow \arg \min_{G^M, r^M, v^M} \mathcal{L}_{coarse}$ 
    // Stage 2: Non-topological Detail Geometry Reconstruction
    11  $\hat{G}_t^M \leftarrow \text{Motion\_Prior\_Init}(\hat{G}_{t-1}^M, G_{t-1}^M, G_t^M);$ 
    12  $\hat{G}_t^M \leftarrow \arg \min_{\hat{G}^M} \mathcal{L}_{detail}$ 
    // Stage 3: Detail Transfer
    13  $G_t^M \leftarrow \arg \min_{G^M} \mathcal{L}_{trans};$ 
    14  $(r_t^M, v_t^M) \leftarrow \arg \min_{r^M, v^M} \mathcal{L}_{nml};$  // Surface align
    // Stage 4: Gaze Tracking
    15  $G_t^{\{L,R\}} \leftarrow \text{ICP}(G_t^M, G_{t-1}^{\{L,R\}});$ 
    16  $r_t^{\{L,R\}} \leftarrow \text{Velocity\_Prior\_Init}(r_{t-1}^{\{L,R\}}, r_{t-1}^M, r_t^M, I_t^{C_0});$ 
    17  $(r_t^{\{L,R\}}, v_t^{\{L,R\}}) \leftarrow \arg \min_{r^{\{L,R\}}, v^{\{L,R\}}} \mathcal{L}_{image};$ 
    18  $(M_t, B_t^{\{L,R\}}) \leftarrow$ 
        Export_Mesh( $G_t^M, r_t^M, v_t^M, G_t^{\{L,R\}}, r_t^{\{L,R\}}, v_t^{\{L,R\}}$ )
19 end

```

and prevent severe overlap between adjacent Gaussian points. We use a scale regularization term to compress the Gaussian ellipsoid to a plane in the direction of the minimum scale and prevent the maximum scale from significantly exceeding the initial value s_{init}^M :

$$\mathcal{L}_{scale} = \sum_{i \in G_0^M} (\min(s_{0,i}^M) + \max(0, \max(s_{0,i}^M) - s_{init,i}^M)). \quad (4)$$

The final loss of the first frame initialization can be summarized as:

$$\mathcal{L}_{init} = \mathcal{L}_{image} + \lambda_{scale} \mathcal{L}_{scale}. \quad (5)$$

4.1.2 Topology-Preserving Coarse Mesh Tracking. In each subsequent frame, we reconstruct G_t^M by deforming the previous Gaussian head G_{t-1}^M . We observe that Gaussian points struggle to perceive global motion, and the capture of head pose cannot be achieved by directly deforming the mesh vertices. We thus divide the coarse head tracking stage into two steps. Firstly, we explicitly model and

calibrate the global motion (i.e., pose) of the whole mesh, and then track the local deformation of each vertex.

Head Pose Representation. We formulate the head pose as a learnable rotation $r_t^M \in \mathbb{R}^6$ described by the 6D representation and a global shift $v_t^M \in \mathbb{R}^3$, where the 6D rotation is the first two columns of a $SO(3)$ matrix and has been considered as a better rotation representation compared to quaternions or Euler angles for its uniqueness and lack of periodicity [Albiero et al. 2021; Hempel et al. 2024]. The 6D rotation can be easily mapped back into $SO(3)$ through the Gram-Schmidt mapping: $R_t^M = f_{GS}(r_t^M)$, and the posed head vertices $\tilde{\mu}_t^M$ at frame t can be obtained by: $\tilde{\mu}_t^M = f_{GS}(r_t^M) \mu_t^M + v_t^M$.

Landmark-Based Pose Calibration. To track head pose, we first utilize the facial priors in the 2D face analyser to roughly align the head pose and give a good start for the subsequent optimization. This pose calibration process is grounded in the observation that the positions of eye corners and the nose tip exhibit significant invariance to head pose and expression change, yet are sufficient to determine the pose roughly. We thus select seven 3D landmarks a_t^M from $\tilde{\mu}_t^M$ to represent these positions and align them with their 2D counterparts across multiple views. The multi-view 2D projection loss between the selected mesh vertices and the estimated landmarks l_t is defined as:

$$\mathcal{L}_{ldm} = \sum_{k=1}^7 (l_{t,k} - \text{proj}(a_{t,k}^M, W_c, K_c)), \quad (6)$$

where $\text{proj}(\cdot)$ project the 3D vertex to the 2D image space by the known camera calibration of view c . Considering the error of the estimated landmarks is significant in the side views, we define head orientation using the normal vector at the nose tip vertex and only optimize over cameras whose orientation lies within a 60° range relative to the head.

Joint Tracking of Pose and Mesh Deformation. After the head pose is roughly determined, we photometrically track the local deformation of each vertex constrained by topological and physical priors. Besides, considering that the landmark-based pose calibration confronts significant errors, especially when the expression is extreme, we also jointly refine the head pose together with the expression dynamics in this step. Specifically, the physical prior losses apply rigid rotation and distance constraints to topologically adjacent vertices:

$$\mathcal{L}_{rot} = \frac{1}{2n_e^M} \sum_{i \in G_t^M} \sum_{j \in \mathcal{K}_i} w_{i,j} \|\hat{q}_{t,j} \hat{q}_{t-1,j}^{-1} - \hat{q}_{t,i} \hat{q}_{t-1,i}^{-1}\|_2, \quad (7)$$

$$\mathcal{L}_{iso} = \frac{1}{2n_e^M} \sum_{i \in G_t^M} \sum_{j \in \mathcal{K}_i} w_{i,j} \|\mu_{0,j} - \mu_{0,i}\|_2 - \|\mu_{t,j} - \mu_{t,i}\|_2, \quad (8)$$

where $\hat{q}$ is the normalized quaternion, and $\mathcal{K}_i$ is the one-ring neighbour of the i -th Gaussian point $G_{t,i}^M$. The weighing factor w is calculated by: $w_{i,j} = \exp(-\lambda_w \|\mu_{0,j} - \mu_{0,i}\|_2^2)$.

The topological prior losses are used to maintain a smooth mesh and avoid serration and interpenetration to improve mesh quality:

$$\mathcal{L}_{pos} = \frac{1}{n_v^M} \sum_{i \in G_t^M} (\mu_{t,i} - \frac{\sum_{j \in \mathcal{K}_i} \mu_{t,j}}{|\mathcal{K}_i|})^2, \quad (9)$$

$$\mathcal{L}_{flat} = \sum_{\theta_i \in E_M} (1 - \cos(\theta_{t,i} - \theta_{0,i})), \quad (10)$$

where $\theta_{t,i}$ is the angle between the faces defined by the topology E_M at frame t . It is worth noting that the topological losses inevitably smooth the mesh and drop details, but the purpose of the current tracking stage is only to obtain a topological mesh that is roughly aligned with the image. We will address the issue of over-smoothing in Sec. 4.3. Overall, the losses in the coarse tracking stage can be formulated as:

$$\mathcal{L}_{coarse} = \mathcal{L}_{image} + \lambda_{rot} \mathcal{L}_{rot} + \lambda_{iso} \mathcal{L}_{iso} + \lambda_{pos} \mathcal{L}_{pos} + \lambda_{flat} \mathcal{L}_{flat}. \quad (11)$$

In this stage, we jointly optimize the head pose ($\mathbf{r}_t^M$ and $\mathbf{v}_t^M$) and the geometric gaussian attributes (μ_t^M and $\mathbf{q}_t^M$).

4.2 Reconstructing Detailed Gaussian Head

Since it is difficult to use a single representation to learn high-precision details while keeping the topology stable, we simultaneously reconstruct a high-precision but non-topological Gaussian head $\hat{G}_t^M$ based on the prior knowledge from the topological head G_t^M to learn geometric details.

4.2.1 Dense Gaussian Head Initialization. Learning detailed geometry needs denser Gaussian points than G_t^M . We thus build a denser Gaussian head at the first frame, and reconstruct detailed geometry frame-by-frame. Following Topo4D [Li et al. 2024a], we initialize $\hat{G}_0^M$ by utilizing the topology prior from the coarse mesh and insert new points into the mesh grids of G_0^M . Specifically, for each original quadrilateral grid, we uniformly insert $N \times N - 4$ vertices into the edges and interior of the grid, and link these vertices with adjacent vertices to form small quadrilateral grids. For each inserted Gaussian point $\hat{G}_{0,i}^M$ that is generated from $\mathcal{F}_i$, where $\mathcal{F}_i$ are the 4 Gaussian points in $G_{0,i}^M$ that forms the quadrilateral grid where $\hat{G}_{0,i}^M$ is located, the Gaussian attributes $\{\hat{\mu}_{t,i}^M \in \mathbb{R}^3, \hat{\mathbf{q}}_{t,i}^M \in \mathbb{R}^4, \hat{\mathbf{s}}_{t,i}^M \in \mathbb{R}^3, \hat{\mathbf{c}}_{t,i}^M \in \mathbb{R}^3, \hat{\sigma}_{t,i}^M \in \mathbb{R}\}$ are initialized by bilinear interpolating the attributes of $\mathcal{F}_i$.

4.2.2 Non-topological Detail Geometry Reconstruction. Inspired by PGSR [Chen et al. 2024], we use a planar-based depth rendering method to render the unbiased Gaussian normal and depth, and apply the single-view and multi-view constraints to optimize detailed geometry. The Gaussian normal map $\mathbf{N}^c$ under view c is calculated by:

$$\mathbf{N}^c = \sum_{i=1} T_i \mathbf{R}_c^T \mathbf{n}_i \alpha_i, T_i = \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (12)$$

where T_i is the accumulated opacity, $\mathbf{R}_c$ is the camera-to-world rotation, and $\mathbf{n}_i$ is the normal direction of the Gaussian point. Subsequently, the unbiased Gaussian depth $L^c(\mathbf{p})$ at image space $\mathbf{p} = [u, v]^T$ can be obtained by:

$$L^c(\mathbf{p}) = \frac{\sum_{i=1} T_i d_i \alpha_i}{\mathbf{N}^c(\mathbf{p}) \mathbf{K}_c^{-1} \tilde{\mathbf{p}}}, d_i = (\mathbf{R}_c^T (\mu_i - T_c))^T (\mathbf{R}_c^T \mathbf{n}_i), \quad (13)$$

where d_i is the Gaussian point's distance to the world-space camera center T_c , and $\tilde{\mathbf{p}}$ is the homogeneous coordinate of $\mathbf{p}$. The edge-aware single-view geometric loss constrains the local consistency

between the rendered depth and normals:

$$\mathcal{L}_{sugeo} = \frac{1}{n_t^c} \sum_{\mathbf{p} \in \mathbf{I}_t^c} (1 - \nabla \mathbf{I}_t^c) \|D_t^c(\mathbf{p}) - \mathbf{N}_t^c(\mathbf{p})\|, \quad (14)$$

$$D_t^c(\mathbf{p}) = \frac{(\mathbf{P}_1 - \mathbf{P}_0) \times (\mathbf{P}_3 - \mathbf{P}_2)}{|(\mathbf{P}_1 - \mathbf{P}_0) \times (\mathbf{P}_3 - \mathbf{P}_2)|}, \quad (15)$$

where n_t^c is the number of head pixels in image $\mathbf{I}_t^c$, $\nabla \mathbf{I}_t^c$ is the normalized image gradient, and $D_t^c(\mathbf{p})$ is the depth-based normal under the local plane assumption calculated by four camera-space 3D points $\mathbf{P}_{\{0,1,2,3\}}$, which are obtained by lifting four adjacent pixels $\mathbf{p}_{\{0,1,2,3\}}$ using their depth $L^c(\mathbf{p}_{\{0,1,2,3\}})$. In addition, we also apply multi-view constraints regarding both geometry and appearance. The geometric loss ensures the multi-view geometric consistency by computing the reprojection error between two near views c_1 and c_2 :

$$\mathcal{L}_{mveo} = \frac{1}{n_t^{c_1}} \sum_{\mathbf{p} \in \mathbf{I}_t^{c_1}} w(\mathbf{p}) \phi(\mathbf{p}), \phi(\mathbf{p}) = \|\mathbf{p} - \mathbf{H}_{c_2 c_1} \mathbf{H}_{c_1 c_2} \mathbf{p}\|, \quad (16)$$

where the homography matrix $\mathbf{H}_{c_1 c_2} = \mathbf{K}_{c_2} (\mathbf{R}_{c_1 c_2} - \frac{T_{c_1 c_2} \mathbf{n}_{c_1}^T}{d_{c_1}}) \mathbf{K}_{c_1}^{-1}$ maps the pixel $\mathbf{p}$ in view c_1 to view c_2 through the relative rotation $\mathbf{R}_{c_1 c_2}$ and translation $T_{c_1 c_2}$. The weight $w(\mathbf{p}) = \exp(-\phi(\mathbf{p})) \cdot \mathbf{1}_{\{\phi(\mathbf{p}) < 1\}}$ regard a pixel is occluded. Besides, the multi-view appearance loss constrains the photometric consistency of the corresponding 7×7 patch between adjacent frames quantized with normalized cross correlation (NCC) [Yoo and Han 2009]:

$$\mathcal{L}_{mvncc} = \frac{1}{n_t^{c_1}} \sum_{\mathbf{p} \in \mathbf{I}_t^{c_1}} w(\mathbf{p}) (1 - \text{NCC}(\mathbf{I}_t^{c_1}(\mathbf{p}), \mathbf{I}_t^{c_2}(\mathbf{H}_{c_1 c_2} \mathbf{p}))). \quad (17)$$

Motion-prior Guidance and Weak Topology-prior Constraint. We observe that the geometric losses can well reconstruct the static head, but are difficult to efficiently reconstruct dynamic head motions. Although $\hat{G}_t^M$ does not strictly maintain the topology, we can let $\hat{G}_t^M$ be softly rigged to G_t^M and utilize the motion prior from the coarse mesh as guidance. Specifically, before optimizing $\hat{G}_t^M$, we drive dense Gaussian points $\hat{G}_{t,i}^M$ by the sparse Gaussian points $G_{t,i}^M$:

$$\hat{\mu}_{t,i}^M = \hat{\mu}_{t-1,i}^M + \sum_{j \in \mathcal{F}_i} b_{i,j} (\mu_{t,j}^M - \mu_{t-1,j}^M), \quad (18)$$

where the motion of $\hat{G}_{t,i}^M$ is initialized by bilinear interpolating the motion of the Gaussian points in $\mathcal{F}_i$ by weight $b_{i,j}$. Besides, we also apply a weak topology constraint to avoid the dense Gaussians deviating too far from their rigged positions:

$$\mathcal{L}_{rig} = \sum_{i \in \hat{G}_t^M} (\|\hat{\mu}_{t,i}^M - \sum_{j \in \mathcal{F}_i} b_{i,j} \mu_{t,j}^M\|_2 - \epsilon)^2, \quad (19)$$

where $\epsilon = 0.005$ is the range of the Gaussian offsets. This term helps to maintain the stability of the Gaussian point density in different facial regions during tracking.

The overall losses for reconstructing the detailed head can be formulated as:

$$\mathcal{L}_{detail} = \mathcal{L}_{image} + \lambda_{sugeo} \mathcal{L}_{sugeo} + \lambda_{mveo} \mathcal{L}_{mveo} + \lambda_{mvncc} \mathcal{L}_{mvncc} + \lambda_{rig} \mathcal{L}_{rig}. \quad (20)$$

In this stage, we optimize all Gaussian attributes of the dense Gaussian head $\hat{G}_t^M$.

4.3 Normal-Guided Geometric Detail Transfer

4.3.1 Multi-View Normal Consistency Regularization. Based on the detailed non-topological Gaussian head $\hat{G}_t^M$, we can transfer its geometric details to the topological coarse Gaussian head G_t^M by multi-view normal consistency constraints. Specifically, we use a differentiable mesh rasterizer $\mathcal{R}_v$ to render the vertex-based normals V_t^c of the posed coarse mesh $(\tilde{\mu}_t^M, E_M)$:

$$V_t^c = \mathcal{R}_v(\tilde{\mu}_t^M, E_M, W_c, K_c), \tilde{\mu}_t^M = f_{GS}(\mathbf{r}_t^M) \mu_t^M + \mathbf{v}_t^M. \quad (21)$$

Then the geometric details can be transferred to G_t^M by the edge-aware normal-consistency loss $\mathcal{L}_{nml}$ that aligns the mesh normals to the detailed Gaussian normals rendered by Eq. 15:

$$\mathcal{L}_{nml} = \frac{1}{n_t^c} \sum_{p \in I_t^c} (1 - \overline{\nabla I_t^c}) \|D_t^c(p) - V_t^c(p)\|. \quad (22)$$

Besides the normal-consistency loss, we also apply the losses in the coarse tracking stage to prevent interpenetration and ensure the photometric consistency of the facial regions. Finally, the losses can be summarized as:

$$\mathcal{L}_{trans} = \mathcal{L}_{coarse} + \lambda_{nml} \mathcal{L}_{nml}. \quad (23)$$

In this stage, we only optimize the geometric-related Gaussian attributes $(\mu_t^M$ and $\mathbf{q}_t^M)$.

4.3.2 Unbiased Surface Alignment. Directly using the mean positions $\tilde{\mu}_t^M$ of Gaussian points as mesh vertices will result in a significant accuracy drop, as the Gaussian points have a volume and there is a gap between the surface formed by $\tilde{\mu}_t^M$ and the real surface. In order to align the extracted mesh with the real surface, Topo4D [Li et al. 2024a] proposed the Gaussian normal expansion method that directly calculates the distance from each Gaussian center to its normal direction ellipsoid surface as the vertex offset. Such an approach is biased, especially when the expression is extreme. Considering that we have the unbiased multi-view normals, we can use the normals to guide the surface alignment. Specifically, before extracting $\tilde{\mu}_t^M$ as the mesh vertices, we apply the normal loss $\mathcal{L}_{nml}$ to optimize the head pose $\mathbf{r}_t^M$ and $\mathbf{v}_t^M$. This simple practice effectively aligns the extracted surface with the real surface. Finally, we can extract the mesh as $M_t := (\tilde{\mu}_t^M, E_M)$.

4.4 Gaze Tracking

With the accurate facial geometry, our next step is to align the eyeball to the mesh and track the gaze. Noting that we mainly focus on gaze tracking and do not involve high-precision eyeball reconstruction. We use a textured eyeball template for different identities. Following the practice of representing the coarse head Gaussian mesh (Sec. 4.1), we represent the eyeball by the Gaussian mesh $G_t^{\{L,R\}}$ where the Gaussian points $G_{t,i}^{\{L,R\}} := \{\mu_{t,i}^{\{L,R\}} \in \mathbb{R}^3, \mathbf{q}_{t,i}^{\{L,R\}} \in \mathbb{R}^4, \mathbf{s}_{t,i}^{\{L,R\}} \in \mathbb{R}^3, \mathbf{c}_{t,i}^{\{L,R\}} \in \mathbb{R}^3, \sigma_{t,i}^{\{L,R\}} \in \mathbb{R}\}$ are rigged to the mesh vertices and the colors are sampled from the template texture map. Similar to the head pose, we formulate the eyeball gaze as the learnable 6D rotation $\mathbf{r}_t^{\{L,R\}} \in \mathbb{R}^6$ and shift $\mathbf{v}_t^{\{L,R\}} \in \mathbb{R}^3$. It's worth noting that the rotation and shift are all in the local head space, which is irrelevant to the head pose, and the neutral gaze means looking straight ahead. This makes it easy to decouple gaze from head pose and helps retarget gaze to other characters.

4.4.1 First Frame Eyeball Pose and Color Fitting. Before tracking gaze, we first initialize the position and gaze of the eyeball in the first frame. To fit the position of the eyeball, we select six vertices around the eyelid in the face mesh $(\tilde{\mu}_t^M, E_M)$ and corresponding vertices on the template eyeball mesh $(V_0^{\{L,R\}}, E_B)$. Then the rigid ICP calculates the transformation T_t^{icp} of fitting the eyeball to the face, which will position the eyeball with neutral gaze. Note that we perform the ICP once at each frame before gaze tracking to ensure the binding of the eyeball and the face. The world-space eyeball vertices can be formulated as:

$$\mu_t^{\{L,R\}} = f_{GS}(\mathbf{r}_t^{\{L,R\}}) T_t^{icp} V_0^{\{L,R\}} + \mathbf{v}_t^{\{L,R\}}, T_t^{icp} = icp(V_0^{\{L,R\}}, \tilde{\mu}_t^M). \quad (24)$$

Since we employ a generic eye template for all performers, it is challenging to optimize the initial gaze before acquiring the eyeball appearance. Instead, we use an off-the-shelf eye landmark detector to estimate the 2D landmarks, and then use a landmark loss similar to Eq. 6 to optimize $\mathbf{r}_0^{\{L,R\}}$ and $\mathbf{v}_0^{\{L,R\}}$ as the initial gaze. After the eyeball is positioned properly, we can fit the appearance of the eyeball. Directly optimizing the color of each Gaussian point may result in inconsistent semantic correspondence. We thus divide the eyeball into three parts (iris, pupil, and sclera), and for each part we optimize a shared color for all Gaussian points in the part. The eyeball color is fixed during the following tracking process. Although the optimized color is not accurate due to light reflection, we observe that it is sufficient for tracking.

4.4.2 Tracking with Velocity and Appearance Prior. It is challenging to track gaze from images because most areas of the eyeball are occluded, and directly optimizing from the limited appearance is unreliable when eyelid occlusion occurs. To reduce ambiguity, we leverage the prior knowledge from 2D gaze estimation methods. Specifically, we use an off-the-shelf generalizable gaze estimation model $\mathcal{G}$ to obtain the world-space $SO(3)$ gaze matrix $\mathbf{g}_t = \mathcal{G}(I_t^{c_0})$ of the front view c_0 . Before tracking each frame, we initialize the gaze by adding the relative estimation $\mathbf{g}_t \mathbf{g}_{t-1}^T$ of adjacent frames as the gaze velocity:

$$\mathbf{R}_t^{\{L,R\}} = \mathbf{g}_t \mathbf{g}_{t-1}^T f_{GS}(\mathbf{r}_t^M)^T f_{GS}(\mathbf{r}_{t-1}^M) f_{GS}(\mathbf{r}_{t-1}^{\{L,R\}}), \quad (25)$$

and the initial gaze $\mathbf{r}_t^{\{L,R\}}$ is the first two columns of $\mathbf{R}_t^{\{L,R\}}$.

Beginning from the estimated initial state, we refine the gaze by inverse rendering. In addition to the eyeballs, the features of the eyelids are crucial for identifying current gaze. Since we have reconstructed the facial expression, we can utilize the geometry and appearance priors of the eyelid for better gaze tracking. Specifically, we concatenate the Gaussian head G_t^M with the Gaussian eyeballs $G_t^{\{L,R\}}$, and render them together to get the rendered eyeballs and eyelids. We optimize the gaze by computing the image loss $\mathcal{L}_{image}$ (Eq. 3) within the eyelid region, taking account of both the posture of the eyeballs and their occlusion relationship with the eyelid. Finally, the 3D eyeballs can be extracted as $B_t^{\{L,R\}} = (\mu_t^{\{L,R\}}, E_B)$.

5 RESULTS

5.1 Experiment Setup

Implementation Details. We implement and test our method on a single NVIDIA 3090 GPU. Our method consists of 4 stages with

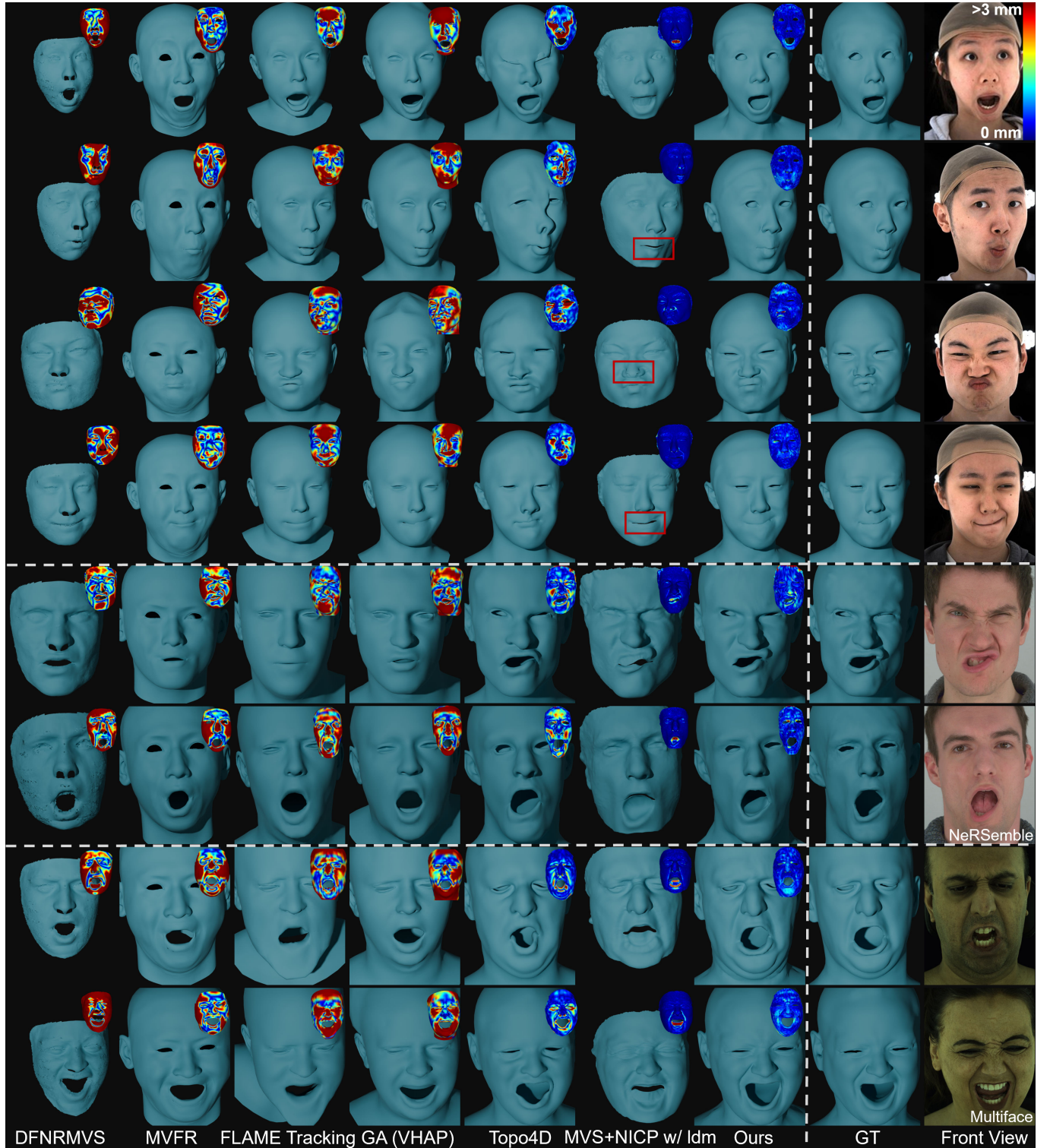


Fig. 4. Qualitative comparison with different face reconstruction methods on three completely different capture systems, including our captured dataset (row 1-4), NeRSemble (row 5-6), and Multiface (row 7-8). The heatmaps show the point-to-surface distance (red indicates ≥ 3 mm), which highlights the geometric error. Our method is robust to various expressions, identities and capture systems. Compared with the scan-free baselines (column 1-5), our method apparently reduces the error, and achieves more robust dense correspondence to extreme expressions than the scan-based registration pipeline (column 6).

Table 1. Quantitative evaluation for different face reconstruction methods on our captured dataset. We measure the percentage of vertices within different error levels and calculate the mean error and variance. We also compare the supported head features of different methods. Compared with FLAME and VHAP, which also support pose and gaze tracking, our method achieves challenging vertex-level topology-consistent tracking, thus significantly reducing the error.

Methods	<0.2mm(%)↑	<0.5mm(%)↑	<1mm(%)↑	<2mm(%)↑	Mean(mm)↓	Med.(mm)↓	Pose	Gaze
DFNRMVS	2.652	6.604	13.040	25.209	5.894	5.235	✓	✗
MVFR	8.699	21.356	40.397	68.355	1.912	1.340	✓	✗
FLAME	8.036	19.738	37.555	64.196	1.968	1.478	✓	✓
GA (VHAP)	7.840	19.379	37.072	63.831	1.903	1.485	✓	✓
Topo4D	12.316	29.731	49.853	68.523	2.233	1.269	✗	✗
Ours	38.891	75.936	93.274	98.017	0.408	0.280	✓	✓

different optimization strategies. 1) The coarse tracking stage can be divided into the initialization of the first frame and the tracking of the following frames. The first frame takes 5000 iterations and needs around 150 seconds for initialization, with $\lambda_{scale} = 10$. The tracking of subsequent frames takes 500 iterations and needs around 30 seconds, where the first 50 iterations are used for landmark-based pose calibration. We set $\lambda_{rot} = 20$, $\lambda_{iso} = 20$, $\lambda_{pos} = 1e3$, and $\lambda_{flat} = 2e - 4$. The image resolution at this stage is 512×375 . 2) The detailed head reconstruction stage takes 3000 iterations for the first frame and 300 iterations for the following frames, which takes around 300 and 30 seconds, respectively. We set $\lambda_{sgeo} = 0.015$, $\lambda_{mgeo} = 0.03$, $\lambda_{moncc} = 0.15$, and $\lambda_{rig} = 10$. The image resolution at this stage is 1024×750 . 3) The detail transfer stage takes 300 iterations and needs around 18 seconds, where the last 50 iterations are used for unbiased surface alignment. We set $\lambda_{nml} = 5$. 4) The gaze tracking stage takes 1000 iterations for the first frame initialization, which needs around 20 seconds. The tracking of other frames takes 200 iterations and needs 10 seconds. The learning rate of learnable Gaussian attributes $\{\mu, q, s, c\}$ are set to $1.6e^{-5}$, $1e^{-3}$, $1e^{-3}$, and 2.5^{-3} , respectively. The learning rate of head pose $\{r^M, v^M\}$ are set to $5e^{-5}$, and the learning rate of eyeball gaze $\{r^{\{L,R\}}, v^{\{L,R\}}\}$ are set to $5e^{-4}$. Our method uses around 8.5 GB of GPU memory in total. We adopt 3DGazeNet [Ververas et al. 2024] to estimate gaze velocity, RTNet [Lin et al. 2021] for face segmentation, and mediapipe to estimate landmarks. All hyperparameters remain the same for all experiments. Our head mesh contains 8280 vertices and 16494 faces. The eyeball contains 770 vertices and 1536 faces.

Datasets. To benchmark methods under simultaneous changes in pose, expression, and gaze, we collected a high-resolution dataset with high-quality facial dynamics using a 24-camera Light Stage. It features 10 identities recorded at 30 FPS and a resolution of 4096×3000 . Performers are asked to freely make facial expressions, rotate their heads and eyes to cover various head poses, extreme expression dynamics and gaze. We collect 30 seconds of data for each identity, amounting to 9000 frames. We use Metashape [Agisoft 2025] to calibrate the cameras and reconstruct scans for computing quantitative metrics. For each subject, we register the neutral mesh of the first frame with Wrap4D [Faceform 2025], which can also be automated by a landmark-based NICP method. Besides the in-house dataset, we also test our method on NeRSemble [Kirschstein et al.

2023] and Multiface [Wuu et al. 2022] to demonstrate the robustness of our method on different identities and capture systems. We use all 16 views from NeRSemble, and select 24 views from Multiface. We resize the images from the two datasets to align the image height with our collected data.

5.2 Face Geometry Comparisons

Baselines. We compare the quality of the reconstructed topological face geometry with five scan-free state-of-the-art multi-view face reconstruction methods from 3 categories: (1) the learning-based method DFNRMVS [Bai et al. 2020] and MVFR [Xiao et al. 2022], (2) the parametric model based FLAME tracking method and VHAP [Qian 2024] (i.e. the tracking methods used in GaussianAvatars [Qian et al. 2024]), and (3) the vertex-level optimization method Topo4D [Li et al. 2024a]. We implement the FLAME-based tracking framework by removing the learnable parameters unrelated to FLAME from VHAP while keeping other settings unchanged. For DFNRMVS, we follow the recommended practice that uses a front view and an oblique side view to produce the best results. Besides the scan-free SOTAs, we also qualitatively compare our method with a traditional MVS and NICP [Amberg et al. 2007] pipeline, where facial landmarks are used.

5.2.1 Quantitative Evaluation. We evaluate facial reconstruction accuracy using mesh-to-scan distances, and the percentage of vertices within different error levels is counted. We only account for the facial region to prevent the influence of scan noise. The results on our collected dataset are reported in Tab. 1. Our method surpasses existing methods in all metrics, achieving submillimeter-level accuracy, and the majority of vertices are within 1 mm precision. Compared to learning-based DFNRMVS and MVFR, our tracking method can cover more extreme expressions and fine-grained geometric details. Attributed to the vertex-level tracking, compared to the parametric-based FLAME tracking and VHAP, our method is not limited by the expressiveness of parametric models and can faithfully reconstruct realistic muscle movements. Although Topo4D also performs vertex-level tracking, it cannot support dynamic head pose. The vertex tracking will fail when the head pose is not accurate, resulting in its poor performance on our collected dataset that includes diverse head rotations. The results also prove that our method can jointly

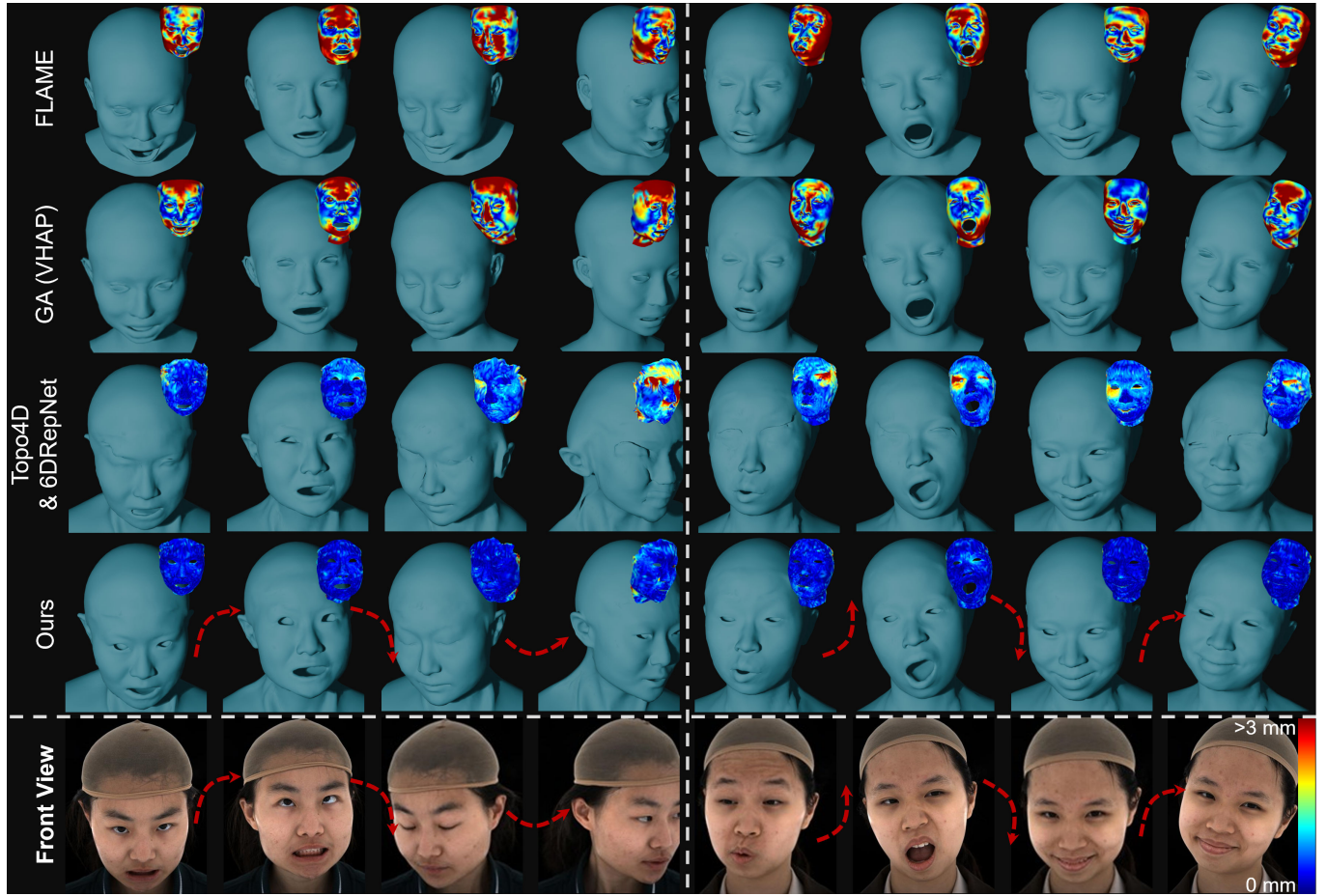


Fig. 5. Qualitative comparison of pose-expression joint tracking on two dynamic sequences with large head rotations. As Topo4D does not support pose tracking, we integrate it with 6DRepNet to estimate the head pose and then perform tracking. The parametric-based tracking methods (FLAME and VHAP) are limited by the expressiveness, making them unable to achieve high-fidelity capture. Compared with the 2D pose estimation plus vertex-based tracking pipeline (Topo4D&6DRepNet), our method accurately tracks the head pose, thus is free from cumulative error and loss of tracking.

capture accurate head pose and facial expression dynamics, as the accurate pose is the foundation of an accurate expression.

5.2.2 Qualitative Evaluation. Fig. 4 shows the visual comparison of reconstructed topological meshes using different methods. We also give example results on the NeRsemble and Multiface datasets. We manually register the template to the scan as the ground truth. Our approach can faithfully reconstruct various extreme expressions and is robust to different races, ages, and capture systems. Due to the proposed novel detail transfer strategy, our method effectively solves the problem of mesh over-smoothing when tracking vertices while maintaining the topology, preserving geometric details. Although traditional MVS and NCP pipelines can accurately fit real surfaces with significantly low geometric errors, the landmark detection often fails when facing extreme expressions, resulting in incorrect semantic correspondence and being easily affected by scan noise. Instead, our method represents facial geometry and appearance priors with dense expressive Gaussian points, thus achieving

more accurate semantic correspondence than sparse landmarks and effectively handling various extreme situations.

5.3 Evaluation of Pose-Expression Joint Tracking

Baselines. Accurate head pose is the foundation for achieving precise head reconstruction. We compare our method with three optimization-based methods. Besides the FLAME tracking framework and VHAP. We also modify Topo4D to support head pose by introducing an off-the-shelf 2D pose estimation method (*i.e.* 6DRepNet [Hempel et al. 2024]) to initialize the mesh pose before tracking the expression dynamics.

Qualitative Evaluation. Fig. 5 displays two dynamic sequences with large head rotations. Although the parametric-based methods can maintain mesh stability even with significant errors in pose and expression, it is limited by low expressiveness and are unable to faithfully reconstruct the facial motion. Importantly, the failure of the Topo4D&6DRepNet pipeline proves that it is challenging

Table 2. Perceptual user study results of different methods, focusing on the temporal consistency and realism of the captured pose, expression and gaze animation. Each aspect is rated from 1 (poor) to 5 (excellent).

Methods	Temporal Consistency $\uparrow$	Pose Realism $\uparrow$	Expression Realism $\uparrow$	Gaze Realism $\uparrow$
Flame Tracking	3.26	3.56	3.27	3.78
GA (VHAP)	3.49	<u>3.69</u>	<u>3.39</u>	3.47
Topo4D & 6DRepNet	<u>4.17</u>	3.24	2.98	-
AGG	-	-	-	3.05
3DGazeNet	-	-	-	3.51
Ours-1 view gaze	-	-	-	<u>4.02</u>
Ours	4.34	4.13	4.53	4.17

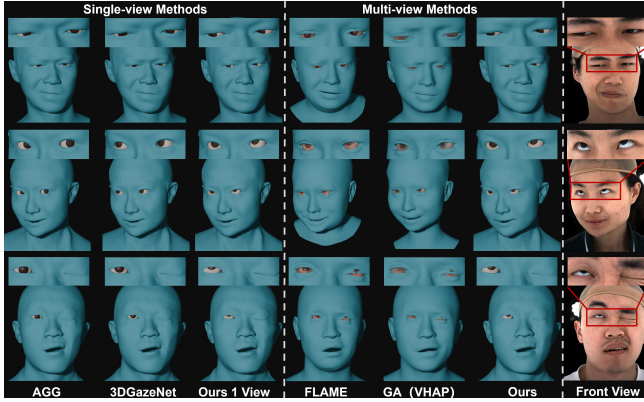


Fig. 6. Qualitative comparison of gaze tracking with monocular and multi-view methods. Compared with baselines, our method is robust to various asymmetric and extreme cases and achieves competitive results with only a single view input.

for the vertex-level optimization methods to maintain stable dense correspondences when the pose is inaccurate. The vertices in the delicate areas, such as the eyes and lips, will easily lose tracking, resulting in significant cumulative errors. The error heat maps show that our method does not exhibit cumulative error when the expression and pose change jointly, which proves that our method can accurately track the head pose, ensuring the stability of the following vertex-level expression optimization process.

5.4 Gaze Tracking Comparisons

Baselines. We compare the visual quality of the gaze tracking with 2 types of methods: 1) the monocular generalizable 2D gaze estimation method AGG [Bao and Lu 2024] and 3DGazeNet [Ververas et al. 2024], and 2) the multi-view FLAME tracking and VHAP. For the 2D methods, we use the facial geometry reconstructed by our method and employ the same eyeball placement method (Sec.24). We use a shared eyeball template texture for better visualization. For VHAP and FLAME tracking, we directly use their facial geometry and estimated textures. In addition to the multi-view results, we also display the monocular gaze tracking results of our method.

Qualitative Evaluation. Fig. 6 demonstrates the gaze tracking results of different methods. Because of the disparities in lighting

Table 3. Efficiency comparisons of optimization-based reconstruction methods per frame.

Methods	Runtime (s)	GPU Memory (GB)	Manual-Free	Scan-Free
MVS+Wrap4D	≈ 600	≈ 3	$\times$	$\times$
MVS+ICP w ldm	≈ 240	≈ 3	$\checkmark$	$\times$
Flame Tracking/VHAP	40.17	6.50	$\checkmark$	$\checkmark$
Topo4D	36.48	1.69	$\checkmark$	$\checkmark$
Ours	83.11	8.39	$\checkmark$	$\checkmark$

conditions in different capture environments, existing generalizable 2D gaze estimation methods have significant errors. Additionally, these methods are unable to handle asymmetric extreme gaze, making it difficult to achieve high-fidelity gaze tracking. Accurate eyelid geometry is crucial for determining gaze status. Contrary to FLAME and VHAP, our method can benefit from the precise expression tracking results and fully utilize geometry and appearance priors to achieve faithful gaze tracking. Importantly, our method can still maintain almost unchanged tracking performance using only a single view.

5.5 User Study of Temporal Stability and Realism

For a comprehensive evaluation, we conducted a perceptual user study to assess the temporal stability and visual realism of the reconstructed results. The study comprises 42 questions, each presenting participants with a randomly selected 7-second video clip alongside corresponding animations reconstructed using different methods. Participants are instructed to evaluate the rendered sequences on a 5-point scale based on their visual plausibility and temporal coherence, using the original front-view video as a reference. 30 participants completed the study, and we report the results in Tab. 2. Our method achieves the highest user preference among all baseline approaches, demonstrating superior temporal stability and producing visually convincing animations.

5.6 Evaluation of Efficiency

We evaluate the computational efficiency of optimization-based reconstruction methods by comparing their peak GPU memory consumption and runtime, with results reported in Tab. 3. All measurements are averaged over 100 consecutive frames on a single RTX 3090 GPU. For the frame-by-frame method, we exclude the first-frame initialization phase and only consider the stable reconstruction runtime. Compared with traditional scan-dependent methods, our approach significantly reduces computational time by eliminating the need for explicit scan reconstruction and manual post-processing, while maintaining competitive output quality. This efficiency gain is attributed to our use of Gaussian-rendered normal maps as geometric supervision, which are less noisy than raw scan data and amenable to incremental optimization. When compared to scan-free methods that produce over-smoothed geometry, our method incurs higher GPU memory and runtime overhead, but delivers substantially improved accuracy in capturing facial dynamics, head pose, and gaze. The additional cost stems primarily from maintaining a dense Gaussian representation for high-fidelity detail transfer. Nevertheless, our method remains computationally feasible for large-scale production environments and is compatible with

Table 4. Quantitative comparisons of ablation studies on our captured dataset towards geometric accuracy. The best and the second-best results are marked with **bold** and underline, respectively.

Methods	<0.2mm(%) ↑	<1mm(%) ↑	Mean(mm) ↓	Median(mm) ↓
w/o coarse tracking	24.905	73.330	1.415	0.700
w/o detail transfer	5.785	38.451	1.497	1.314
w/o surface align	19.309	81.555	0.679	0.555
w/o densification	35.649	92.310	0.438	0.309
w/o motion prior	38.401	91.209	0.453	<u>0.288</u>
w/o $\mathcal{L}_{rig}$	<u>37.687</u>	<u>92.677</u>	<u>0.424</u>	0.291
4 views	27.653	81.376	0.670	0.435
8 views	33.276	89.432	0.494	0.342
16 views	37.300	92.424	0.429	0.294
24 views	38.891	93.274	0.408	0.280

most consumer-grade GPUs. Overall, our framework achieves a favourable balance between joint capture accuracy, geometric detail quality, and computational cost, making it a practical solution for digital asset production.

5.7 Ablation Study

To validate our design choices for the proposed multi-stage framework, we conduct extensive ablation studies by altering the involved key modules. **(1)** We study the necessity of the coarse tracking stage (Sec. 4.1) and the detail transfer stage (Sec. 4.3). **(2)** We study the priors used for promoting the dynamic reconstruction of the unstructured detailed Gaussian head (Sec. 4.2), including the topology-prior guided densification, the motion-prior used for driving dense Gaussians, and the weak topology-prior regularization term. **(3)** We study the effect of the velocity prior and the appearance prior used for gaze tracking (Sec. 4.4). **(4)** We also investigate the influence of the input view number. The statistics of the ablation study are shown in Tab. 4, and the visual examples are shown in Fig. 7, where the differences of the Gaussian normal map (computed by Eq. 15) used for detail transferring are highlighted by the red box.

Coarse-to-Fine Strategy. We skip the coarse tracking stage (noted as w/o coarse tracking), and directly optimize the non-topological dense Gaussian head for detail transferring. This design brings two negative impacts: 1) the subsequent detailed geometry reconstruction process lacks the topology and motion priors provided by the coarse mesh, leading to geometric artifacts, and 2) the facial semantics of the topological Gaussian head are not aligned, resulting in a loss of dense correspondence between Gaussian points and facial regions after detail transferring. All metrics are significantly decreased, and as is shown in Fig. 7 (a), the tracking of the mouth area is lost.

Normal-guided Detail Transfer. We skip the detail transfer stage (noted as w/o detail transfer). This variant causes the over-smoothing problem with more than 60% of the vertices being out of the 1mm error range. Meanwhile, as shown in Fig. 7 (b), using unbiased normals for geometric constraints helps maintain the tracking stability, and the absence of this step can lead to significant cumulative errors, especially with large head pose variations.

Unbiased Surface Alignment. We remove the unbiased surface alignment process (noted as w/o surface align). Although this step

does not affect tracking, the average geometric error increases from 0.408 mm to 0.679 mm with only 5.785% of vertices within the 0.2 mm error. This degradation arises from the gap between the Gaussian surface and the real surface caused by Gaussian point volume, and aligning the Gaussian geometry by unbiased normals can effectively remove this part of the error.

Topology-guided Gaussian Densification. We do not build a dense Gaussian head to learn the detailed geometry, but use the same number of points as the coarse Gaussian mesh (noted as w/o densification). As shown in Fig. 7 (d), the expressiveness of the Gaussian mesh decreases as the density of Gaussian points decreases, resulting in blurry facial areas and affecting the transfer of geometric details to the mesh.

Motion Prior Guidance. We do not use the motion of the coarse Gaussian mesh to drive the dense Gaussian mesh (noted as w/o motion prior). As shown in Fig. 7 (e), unreasonable protrusions appear in the rendered Gaussian normal map and are reflected on the mesh. This validates the proposed motion prior guidance provides good initialization for dense Gaussian points and effectively reduces the difficulty of learning dynamic detail geometry.

Weak Topology-prior Regularization. In this variant, we remove the $\mathcal{L}_{rig}$ (noted as w/o $\mathcal{L}_{rig}$). Fig. 7 (f) shows that removing the geometric constraints from the coarse mesh, the texture-less areas, such as the cheek, tend to be underfitting, displaying artifacts on smooth surfaces. The proposed $\mathcal{L}_{rig}$ can effectively control the distribution of the dynamic Gaussian points on different facial regions.

View Number. We decrease the number of input views to study the effect of our method under sparser captures. As shown in Tab. 4, the performance of our method with 16 views does not drop significantly compared with 24 views. Even with only 4 views input, our method outperforms the baselines in all metrics. Fig. 7 (g) displays that when the number of views decreases, our method can still effectively track and accurately reconstruct most of the facial regions, but the weak observation around the wrinkles leads to geometric degradation. This study validates our method's potential for supporting a lightweight capturing system.

Gaze Velocity Prior. As shown in Fig. 7 (h), after removing the velocity prior-based gaze initialization, the tracking loses when most of the pupils are obstructed. The prior knowledge from 2D gaze estimation models can provide a reliable optimization direction in extreme situations, alleviating the optimization ambiguity caused by insufficient observation.

Eyelid Appearance Prior. We render the Gaussian eyeballs without the Gaussian head and only compute the image loss within eyeball regions when tracking gaze. Fig. 7 (i) displays that simply optimizing the eyeball appearance will lead to ambiguity caused by limited observations as well as interpenetration between eyelids and eyeballs. Our well-designed multi-stage optimization framework allows gaze tracking to be established on accurate facial geometry, fully utilizing the eyelid contour and appearance prior derived from the Gaussian mesh.

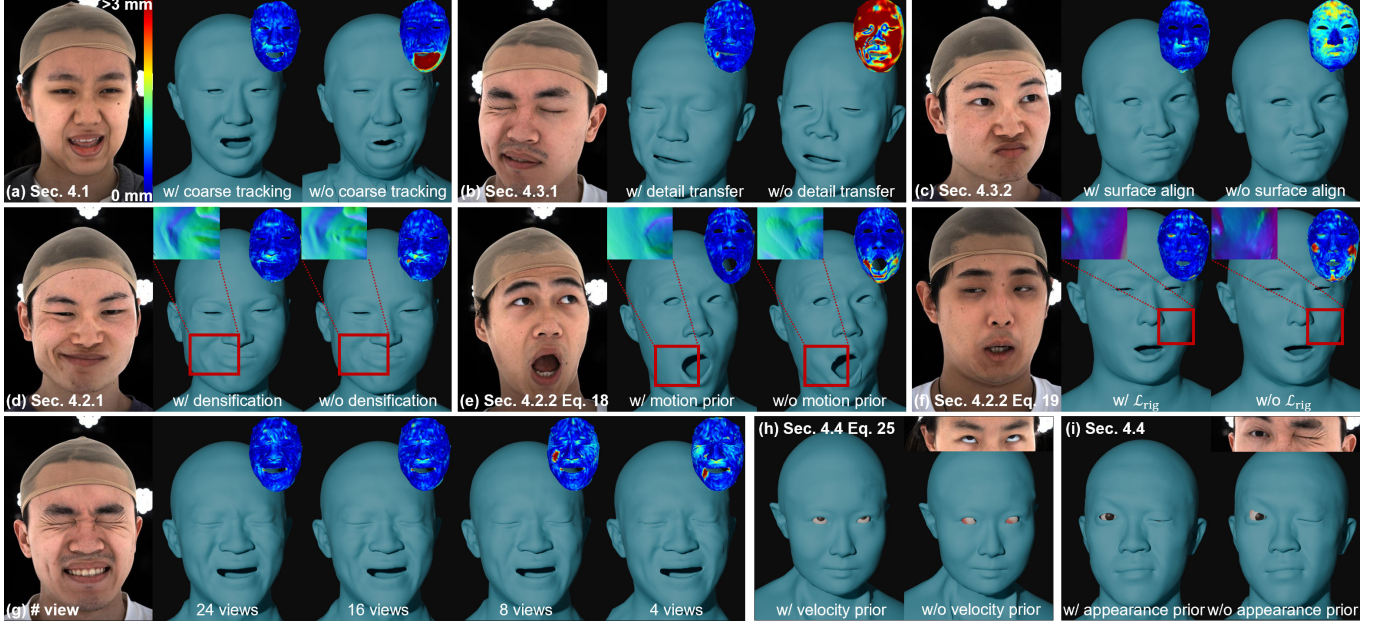


Fig. 7. Visualization of the reconstructed geometry of different ablation variants with heatmaps indicating the geometric error. We highlight the rendered Gaussian normal map (computed by Eq. 15) in the red box. (a) skip the coarse tracking stage, (b) skip the detail transferring stage, (c) directly extract geometry without using normals to align surface, (d) do not densify the Gaussian points when initializing the dense Gaussian head, (e) do not drive the dense Gaussian head by the motion of the coarse Gaussian head, (f) remove $\mathcal{L}_{rig}$, (g) decrease the input view number, (h) do not use 2D gaze prior for gaze initialization, (i) remove the facial expression Gaussian points when tracking gaze.

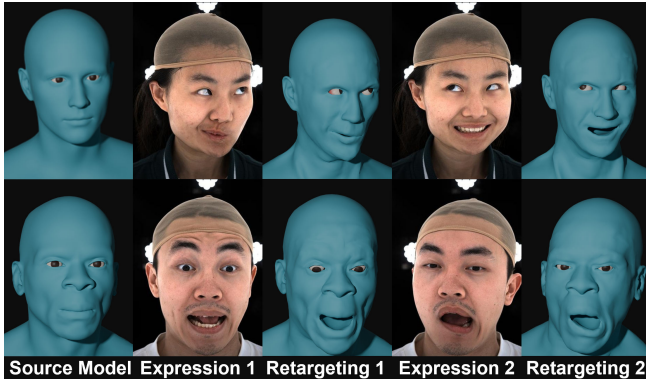


Fig. 8. Examples of retargeting the captured pose, expression and gaze to other characters.

6 APPLICATION: RETARGETING

Facial performance retargeting plays a crucial role in modern digital character animation, enabling the seamless transfer of expressive performances across characters while preserving emotional subtleties and stylistic consistency. The reconstructed pose and facial expression can be directly retargeted to any digital character by applying the corresponding vertex offsets to a neutral expression mesh. Furthermore, since our gaze representation is formulated as a head-space rotation, it can be easily transferred by setting the eyeball rotation after rigidly binding it to the target mesh via ICP.

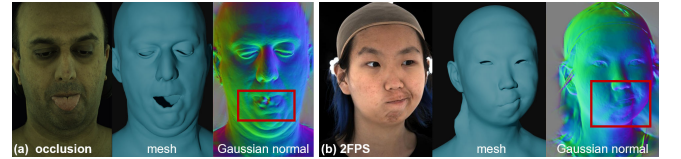


Fig. 9. Failure cases. (a) Severe occlusion causes the loss of tracking. (b) Geometric blur at extreme low frame rate (2FPS).

Fig. 8 demonstrates the example results and displays the practicality of our approach to be integrated into existing industrial 3D asset production pipelines.

7 DISCUSSIONS

Limitations. Although our scan-free method can efficiently capture high-precision 4D facial expression dynamics, head pose and gaze, it has some limitations, as is shown in Fig. 9. (1) Our method fails when heavy overlapping occurs, e.g. the stuck-out tongue covering the lips. Such a scenario is hard to handle by tracking and usually requires manual operation in the industry. (2) Unlike the static reconstruction methods [Fyfe et al. 2017], our method is based on frame-by-frame tracking, so the input captures need to be temporally synchronized, and the continuity between frames is supposed to be sufficient for tracking. We observe that when the frame rate is extremely low ($< 7\text{FPS}$), significant skips between frames will lead to tracking instability and error accumulation. (3) To ensure stable and unbiased tracking, our pipeline requires the first frame to

be a neutral expression with exposed pupils. This initial constraint provides the initial Gaussian set with reliable geometry and appearance priors specific to the subject, which are crucial for subsequent reconstruction.

Future Work. Our method is designed for multi-view facial capture, leveraging detailed normal maps derived from multi-view optimization to bypass the traditional scan reconstruction process while preserving geometric details. Given the rapid advancement of end-to-end visual foundation models [Wang et al. 2025, 2024a], a promising future direction involves integrating large-scale visual priors to accelerate the generation of high-quality surface normals. Such integration could further extend the applicability of our framework to monocular or in-the-wild scenarios through generative priors. Additionally, while our approach enables simultaneous tracking of head pose, detailed facial dynamics, and gaze, it currently does not model internal components such as the teeth and tongue [Rasras et al. 2024; Velinov et al. 2018], nor does it account for hair dynamics [Beeler et al. 2012; Winberg et al. 2022; Zakharov et al. 2024]. The present eye model also remains simplified. Our framework is open to the integration of these components. Another valuable research direction is to incorporate oral cavity, hair, and skull structures into the tracking framework and to embed more sophisticated anatomical and physical constraints for improved realism and robustness.

8 CONCLUSION

We have presented a scan-free framework for simultaneously capturing the topology-consistent head pose, detailed facial expression dynamics and gaze from calibrated multi-view videos. At the core of our approach is a carefully designed four-stage tracking pipeline, where each stage is guided by priors either from 2D models or outputs of preceding stages to ensure robust and coherent tracking. To address the common issue of mesh over-smoothing in existing scan-free 4D reconstruction methods, we decouple the tasks of geometric detail preservation and topological stability using two complementary Gaussian representations. We track a coarse topological mesh to maintain facial structure, which in turn uses its reliable priors to guide the reconstruction of a non-topological Gaussian set for capturing high-frequency geometry and appearance details. A novel normal-guided detail transfer strategy is proposed to effectively integrate the unstructured geometric details from the non-topological Gaussian set into the topological mesh without violating its topology. For accurate head pose and gaze capture, we employ a two-step optimization strategy: initialization with coarse 2D model-based priors, followed by refinement using dense Gaussian-derived priors at the vertex and pixel levels. The 2D priors offer reliable initialization and stabilize the tracking process, while the fine-grained Gaussian priors resolve ambiguities and inaccuracies in the coarse estimates, enabling precise tracking and reducing both cumulative and mutual errors. Our experimental evaluation showcases that the proposed method not only surpasses existing counterparts but also represents a significant advancement beyond the capabilities of scan-free topology-consistent 4D reconstruction.

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